Nebraska Public Power District

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10 CFR 50.73

NLS2025038
June 06, 2025

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2025-001-00
Cooper Nuclear Station, Docket No. 50-298, Renewed License No. DPR-46

The purpose of this correspondence is to forward Licensee Event Report 2025-001-00.

This letter does not contain regulatory commitments.

Sincerely,

Khalil Dia
Site Vice President

/sn

Attachment: Licensee Event Report 2025-001-00

cc: Regional Administrator w/attachment
USNRC - Region IV

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

Senior Resident Inspector w/attachment
USNRC - CNS

SRAB Administrator w/attachment

NPG Distribution w/attachment

INPO Records Center w/attachment
via IRIS entry

SORC Chairman w/attachment

CNS Records w/attachment

NRC FORM 366 (04-02-2024)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Cooper Nuclear Station	☒ 050 ☐ 052	298	1 of 5

4. Title
Inoperable Diesel Generator Jacket Water Pump Results in Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
04	08	2025	2025	001	00	06	06	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Linda Dewhirst, Regulatory Affairs and Compliance Manager	(402) 825-5416

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	DG	P	A180	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)
On April 8, 2025, with Cooper Nuclear Station (CNS) reactor in Mode 1, Power Operation, scheduled monthly surveillance testing of Diesel Generator (DG) 1 was being performed. During this test, a low jacket water (DGJW) temperature alarm was received. Initial investigation identified that the DGJW pump may not be providing sufficient cooling water to the DG. DG1 was subsequently secured and declared inoperable.

During disassembly of the DGJW pump on April 9, 2025, it was discovered that the pump impeller had slipped on the shaft. Repairs were made, subsequent testing performed satisfactorily, and Operations personnel declared DG1 operable.

Investigation into the pump impeller failure identified the direct cause to be that the impeller to shaft fit with an inadequate fit during pump seal replacement on February 5, 2025. DG1 was determined to have been potentially inoperable from as early as February 5, 2025 until April 8, 2025, resulting in a past operation or condition prohibited by Technical Specifications and a condition that could have prevented the fulfillment of a safety function. Corrective actions include purchase of new impellers with correct tolerances and revision to Preventative Maintenance and Vendor Manual work instructions.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Cooper Nuclear Station	05000-298	2025	001	00

NARRATIVE

PLANT STATUS

Cooper Nuclear Station (CNS) was in Mode 1, Power Operation, at 100% power at the time the condition was discovered on April 8, 2025. Diesel Generator (DG) 1 was declared inoperable due to failure of the Diesel Generator Jacket Water (DGJW) [EIIS:LB] pump [EIIS:P] during the monthly operability test.

BACKGROUND

The purpose of the standby (emergency) Alternating Current (AC) power system [EIIS:EK] is to provide a single failure proof source of on-site AC power adequate for maintaining the safe shutdown of the reactor following abnormal operational transients and postulated accidents. This system consists of two independent AC power sources, the Emergency Diesel Generators (EDG) [EIIS:DG].

Each DG shall be capable of automatic start at any time and capable of continued operation at rated load, voltage, and frequency until manually stopped.

During normal plant operations both DGs are in standby. A DG starts automatically on a loss of coolant accident (LOCA) signal (i.e., low reactor water level signal or high drywell pressure) or on loss of voltage on a critical bus or Loss of Offsite Power (LOOP) event. The DG automatically connects to its respective bus after off-site power is tripped as a consequence of critical bus loss of voltage or degraded voltage.

Each DG has an associated DGJW subsystem. The DGJW subsystem is a closed recirculating water circuit containing treated demineralized water for engine cooling. It consists of a standpipe, an engine driven pump, a heat exchanger, a bypass pump, a heater and the associated piping and instrumentation. The DGJW subsystem is required to be operable to support the associated DG operability.

CNS Technical Specifications (TS) require that two EDGs be operable when the plant is in Modes 1, 2, or 3, and that one DG be operable when the plant is in Modes 4 or 5.

EVENT DESCRIPTION

On April 8, 2025, with CNS reactor in Mode 1, Power Operation, scheduled monthly surveillance testing of DG1 was being performed. The engine started and loaded as expected per procedure, however, at full load, the jacket water low temperature alarm [EIIS:TA] was received. Upon reading local DG1 jacket water temperature indication, the jacket water entering the engine was approximately 82 degrees F and the outlet temperature was reading approximately 182 degrees F. Expected condition is for the inlet and outlet jacket water temperatures to converge between 162-166 degrees F.

Operators investigating the alarm noticed the discharge pressure from the engine driven jacket water pump was reading 5psi versus the expected 25-30psi. Operations had previously recorded the pump discharge pressure at approximately 30psi when read shortly before closing generator output breaker and adding load. The combination of diverging jacket water temperatures and low discharge pressure was evidence of low jacket water flow. Due to insufficient cooling to the engine, DG1 was manually secured. The engine ran approximately 90 minutes total and 30-60 minutes with degraded jacket water flow. DG1 was subsequently declared inoperable.

The engine driven jacket water pump was disassembled on April 9, 2025, and it was discovered the impeller spun freely on the shaft and the inner diameter had wear damage from the shaft rotation. The impeller nut and drake nut were found tight. No damage was found to the pump shaft, wear rings, or the mechanical seal.

The damaged impeller was sent to a pump vendor for fabrication and installation of a sleeve within the impeller bore. After machining and assembly, the sleeved impeller was returned and re-installed on the original pump shaft using new vendor work instructions. With the engine driven jacket water pump restored, acceptance test runs were completed satisfactory and DG1 was returned to operable status on April 12, 2025.

During the subsequent cause evaluation, it was identified that the jacket water pump impeller had been removed and re-installed to support mechanical seal replacement on February 5, 2025, using instructions provided by the pump manufacturer. The investigation identified that the impeller re-installation failed to establish an adequate interference fit which allowed the impeller to eventually come loose from the shaft. This was due to the original impeller being manufactured out of tolerance and the assembly instructions being based on the specified dimensions.

A review determined DG1 to have been potentially inoperable from the start of the February maintenance window on February 3, 2025, at 22:30 CST until the jacket water pump was repaired, tested, and DG1 declared operable on April 12, 2025, at 08:20 CST. Technical Specification 3.8.1, AC Sources-Operating, requires the plant to be shutdown if an inoperable DG is not restored to operable status within seven days. Since DG1 was inoperable for greater than seven days without completing a reactor shutdown, this represents a condition which was prohibited by Technical Specifications.

A review of redundant equipment operability during the period that DG1 was potentially inoperable was conducted. DG2 was identified as being inoperable in multiple instances; Station Start-up Service Transformer (SSST) was identified as being inoperable in one instance, and Emergency Station Service Transformer (ESST) was not identified as being inoperable. DG1, DG2, and the SSST were not all inoperable at the same time.

Of the instances where DG2 was inoperable, one exceeded the two-hour required action for Limiting Condition for Operation (LCO) 3.8.1 Condition E – Two DGs inoperable, to restore one DG to operable status and subsequent required action for LCO 3.8.1 Condition F – Required action and associated completion time of Condition A, B, C, D or E not met, to be in Mode 3 in 12 hours. This instance represents a condition prohibited by Technical Specifications and represents a condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident.

The SSST was inoperable concurrent with DG1 for 92.33 hours which exceeds the 24 hour required action for LCO 3.8.1 Condition D – One offsite circuit inoperable and one DG inoperable, to restore either the offsite circuit or affected DG to operable status, and subsequent required action for LCO 3.8.1 Condition F – Required action and associated completion time of Condition A, B, C, D or E not met, to be in Mode 3 in 12 hours. This represents a condition prohibited by Technical Specifications.

BASIS FOR REPORT

This event is being reported per 10 CFR 50.73(a)(2)(i)(B) as "any operation or condition which was prohibited by Technical Specifications" and 10 CFR 50.73(a)(2)(v) as "any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to ...(D) Mitigate the consequences of an accident."

SAFETY SIGNIFICANCE

The key safety consequences of this event include the loss of the ability of DG1 to automatically provide emergency AC power to energize the Division 1 4160 VAC bus during either a LOCA or LOOP event.

The safety significance associated with the DG1 Jacket Water Pump failure is low based on DG2 not being impacted, except as discussed below, and would have performed functions provided by DG1 using redundant Division 2 systems. For a LOCA event, the SSST and ESST were capable of performing the functions provided by DG1. The supplemental DG was not impacted and could also be used to power DG1 loads that are required to mitigate associated LOOP events.

DG2 was removed from service for both planned and unplanned maintenance during the period of time when DG1 was vulnerable to the DGJW pump failure. This configuration resulted in concurrent unavailability of both EDGs, and the corresponding inability to automatically provide emergency onsite AC power. This significance is also judged to be low based on the small exposure time in which both EDGs were unavailable and that the supplemental DG remained available to provide power to loads required to mitigate a LOOP event. The SSST and ESST were available during the times that both EDGs were unavailable.

The SSST was removed from service for planned maintenance during the period of time when DG1 was vulnerable to the DGJW pump failure. This configuration resulted in concurrent unavailability of both DG1 and the SSST, and the corresponding inability of either source to provide power to Division 1 4160 VAC bus during a LOCA event. This significance is also judged to be very low based on the small exposure time in which DG1 and the SSST were unavailable and that the ESST was available with redundant capability to provide power to the Division 1 4160 VAC bus.

This event is considered a Safety System Functional Failure (SSFF).

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CAUSES

The direct cause of the DGJW pump was inadequate interference fit of the impeller onto the shaft. Causal factors were:

- Original manufacturer machining of the impeller tolerances were found to be outside bore dimensional specification.
- Original vendor work instructions were of insufficient detail to ensure repeatable impeller assembly.

CORRECTIVE ACTIONS

- The damaged DG1 DGJW impeller was sent to a pump vendor for fabrication and installation of a sleeve within the impeller bore. After machining and assembly, the sleeved impeller was returned and re-installed on the original pump shaft using new vendor work instructions.
- CNS will obtain replacement impellers manufactured to the original design specifications.
- Preventative maintenance and vendor manual work instructions will be revised to validate impeller bore critical dimensions and shaft placement in accordance with vendor guidance.

PREVIOUS EVENTS

A review of CNS licensee event reports for the last three years was performed. There were no previous occurrences involving a similar failure to that discussed in this report.